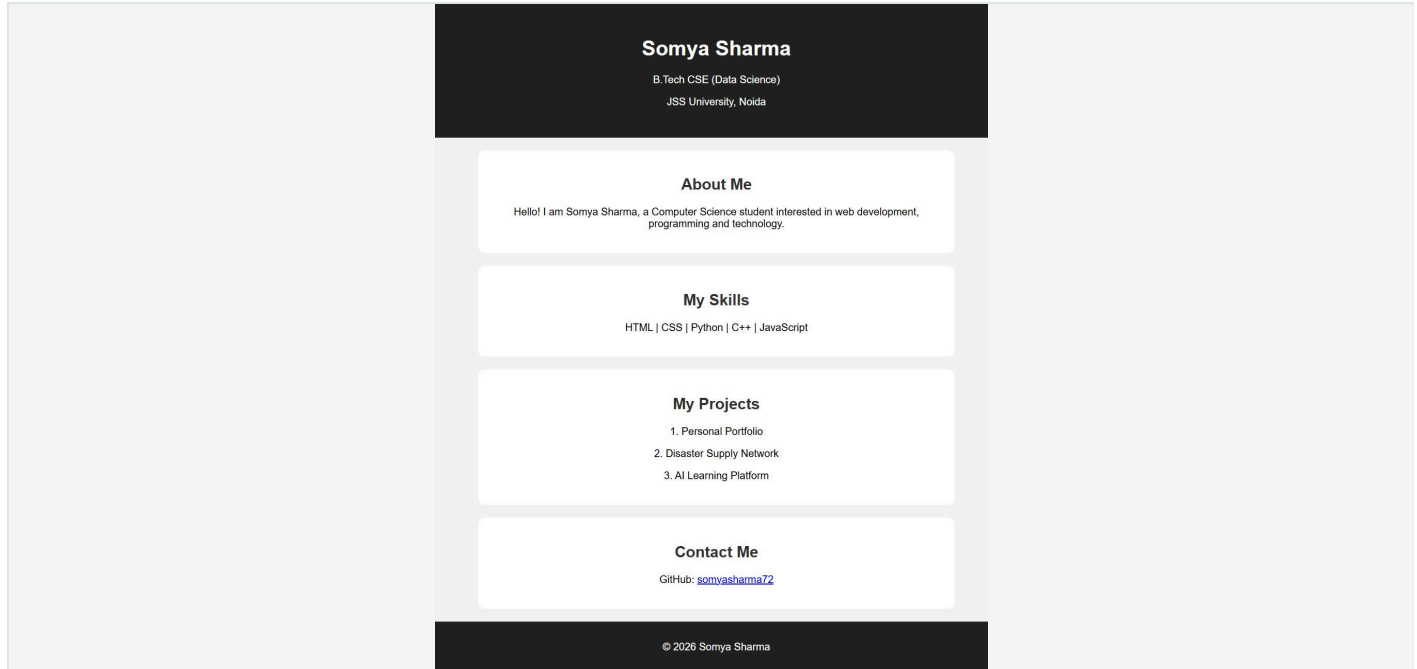


EXPERIMENT 01 OF 06

Personal Profile & GitHub Pages

Profile page with notes for repository setup, branching, merging, and static-site hosting.

Published page: somyasharma72.github.io/college_experiments/experiment1/



Objective and page contents

Present the student's profile, skills, projects, and contact details. The experiment also explains how GitHub stores the project and how GitHub Pages can publish it.

Resources: HTML, CSS, VS Code, Git, GitHub, a web browser, and GitHub Pages.

Project structure

```
experiment1/  
└─ index.html
```

Capture: deployed page returned HTTP 200 on 9 Oct 2026.

Git tasks to perform manually

1. Create a GitHub repository, connect this folder, commit the files, and push the default branch.
2. Create and push a feature branch; make and commit a change.
3. Open a pull request, review it, and merge it into the default branch.
4. Configure Settings > Pages to publish the default branch's root folder; wait for the build and check its URL.

Not claimed as completed: repository creation, branch work, merge, and deployment actions must be performed and verified by the student.

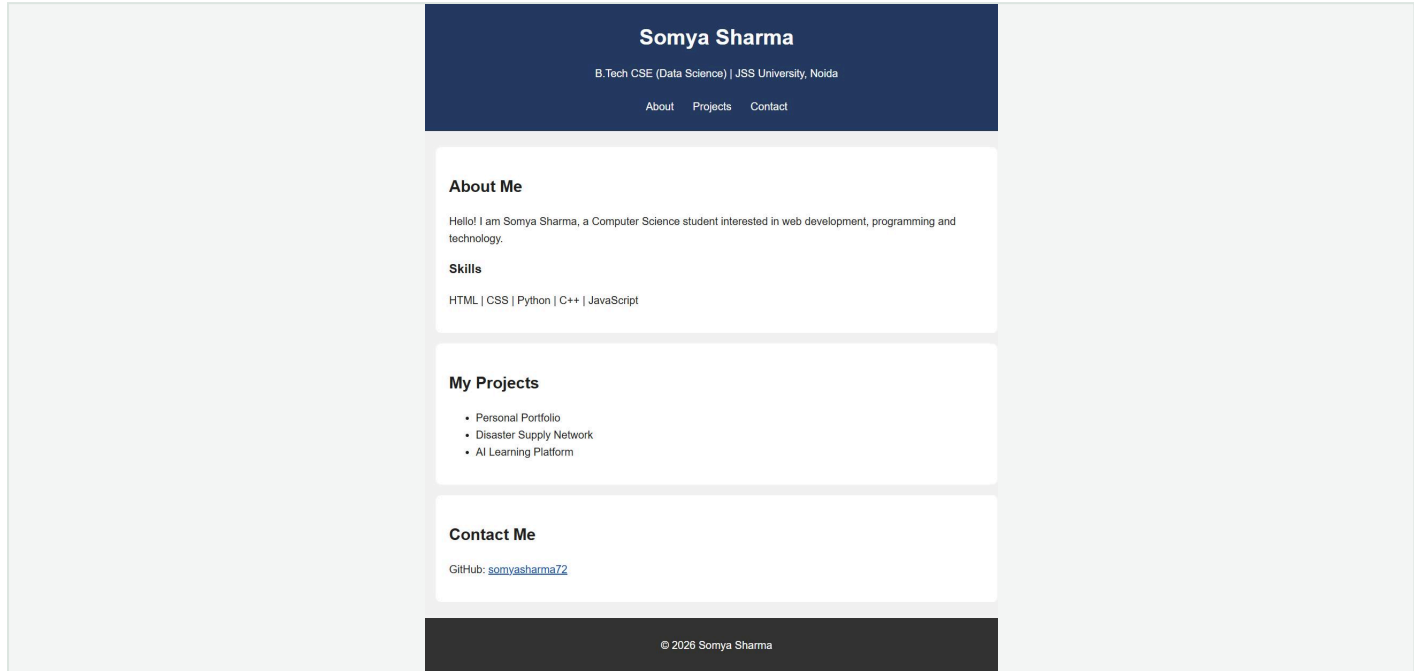
Screenshot reflects the published page at the time of capture. Local source may differ until pushed and deployed.

EXPERIMENT 02 OF 06

Personal Portfolio

A responsive student portfolio with section navigation, About, Projects, and Contact.

Published page: somyasharma72.github.io/college_experiments/portfolio/



Objective and page contents

Introduce the student, show a small skills list and project list, and provide navigation links to the page sections.

Resources: HTML5 and a linked CSS stylesheet.

Project structure

```
portfolio/  
├── index.html  
└── style.css
```

Capture: deployed page returned HTTP 200 on 9 Oct 2026.

Implementation excerpt

```
<nav aria-label="Main navigation">  
  <a href="#about">About</a>  
  <a href="#projects">Projects</a>  
  <a href="#contact">Contact</a>  
</nav>  
<main>  
  <section id="about">...</section>  
  <section id="projects">...</section>  
  <section id="contact">...</section>  
</main>
```

The navigation targets the matching section IDs. The stylesheet provides a centered layout and a small-screen spacing adjustment.

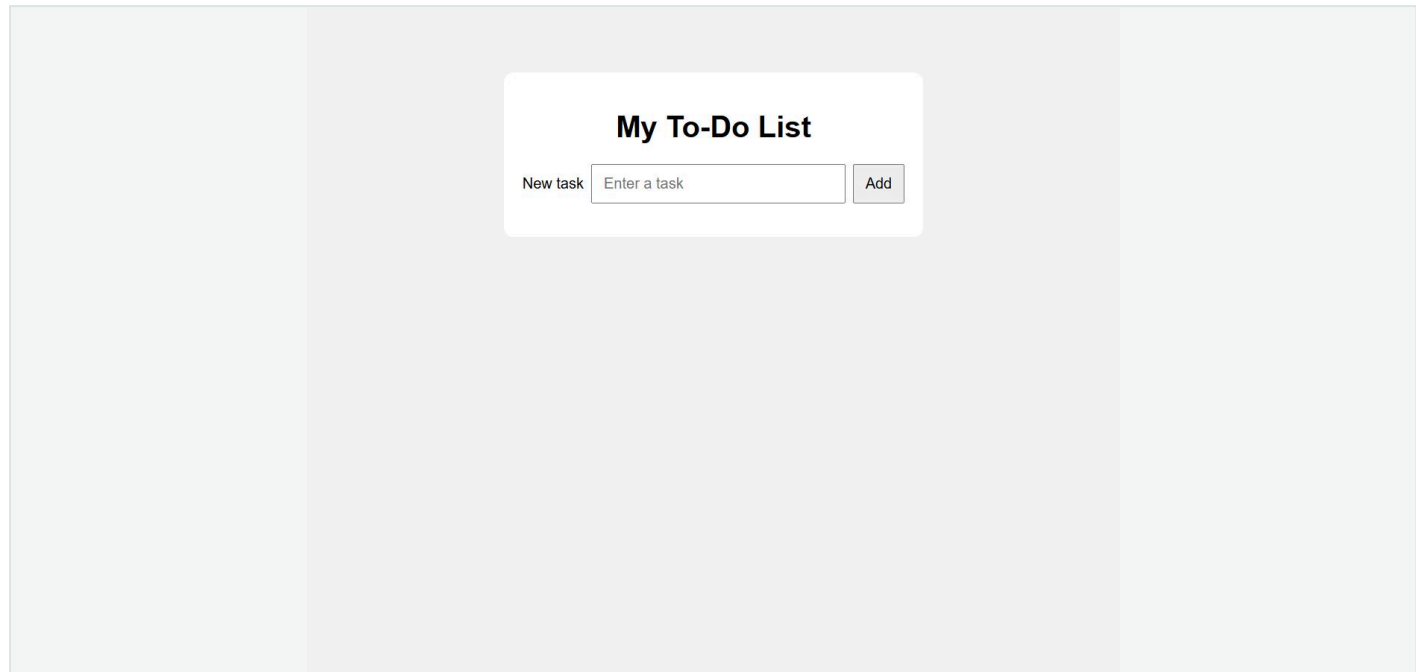
Screenshot reflects the published page at the time of capture. Local source may differ until pushed and deployed.

EXPERIMENT 03 OF 06

To-Do List

A small browser-based task list for adding, completing, and deleting tasks.

Published page: somyasharma72.github.io/college_experiments/to-do%20list/



Objective and page contents

Accept a task, append it to a list, toggle its completed state, and remove it when no longer needed. Empty task text is ignored.

Resources: HTML, CSS, and browser JavaScript.

Project structure

```
to-do list/  
└─ index.html  
(styles and script are inline)
```

Capture: deployed page returned HTTP 200 on 9 Oct 2026.

Implementation excerpt

```
taskForm.addEventListener("submit", function (event) {  
  event.preventDefault();  
  const task = taskInput.value.trim();  
  if (!task) return;  
  // Create task text, Complete/Undo and Delete buttons.  
  taskList.appendChild(item);  
});
```

Task text is inserted with `textContent`; the action buttons work with keyboard and mouse.

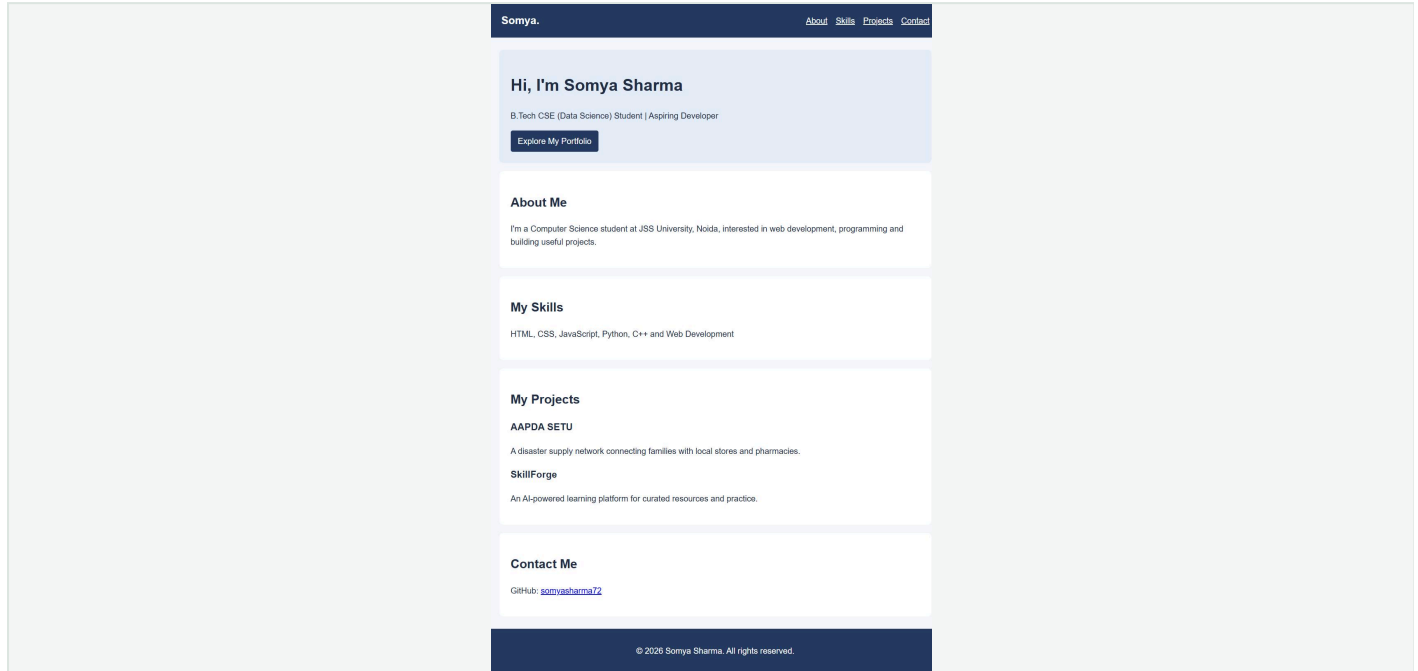
Screenshot reflects the published page at the time of capture. Local source may differ until pushed and deployed.

EXPERIMENT 04 OF 06

SEO & Accessibility

Portfolio page improvements for search metadata, semantic structure, keyboard access, and mobile layouts.

Published page: somyasharma72.github.io/college_experiments/seo%20experiment/



Checks and improvements

- English document language, descriptive title and meta description.
- Header, labeled navigation, main content, sections, and footer.
- Heading order, valid section links, and visible keyboard focus.
- Responsive CSS and reduced-motion preference support.
- No images or form inputs are used on this page.

Project structure

```
seo experiment/  
├── index.html  
└── style.css
```

Resources: HTML5 and CSS3. Deployed page returned HTTP 200 on 9 Oct 2026.

Implementation excerpt

```
<meta name="description"  
  content="Somya Sharma is a B.Tech Computer Science student...">  
<header>...<nav aria-label="Main navigation">...</nav></header>  
<main id="top">  
  <section id="about">...</section>  
</main>
```

The canonical URL was intentionally removed because the original pointed to `example.com`.

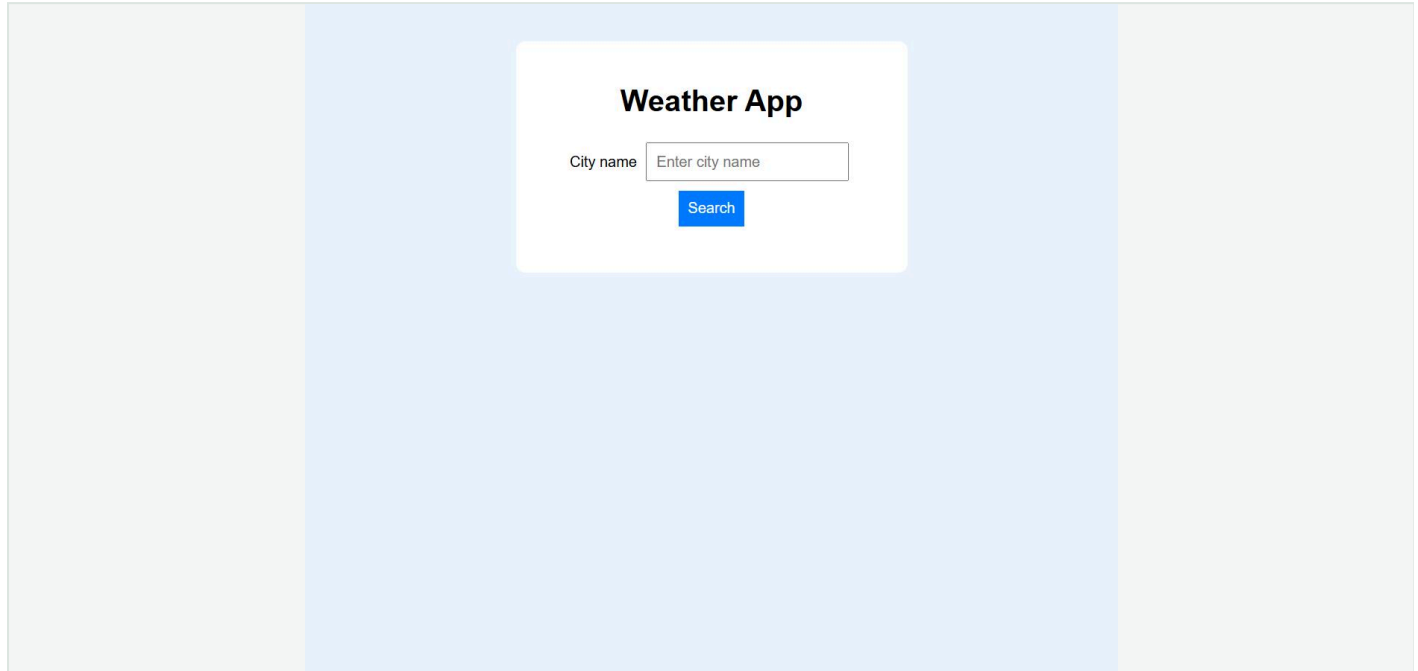
Screenshot reflects the published page at the time of capture. Local source may differ until pushed and deployed.

EXPERIMENT 05 OF 06

Weather Application

City search uses Open-Meteo geocoding followed by its forecast API to show current conditions.

Published page: somyasharma72.github.io/college_experiments/weather/



Objective and behavior

Search by city name, show a loading message, handle unknown cities and API/network errors, and display temperature, humidity, and wind speed when data is returned.

Resources: HTML, CSS, JavaScript Fetch API, and Open-Meteo public endpoints.

Project structure

```
weather/  
├── index.html  
(styles and script are inline)
```

Capture: deployed page returned HTTP 200 on 9 Oct 2026.

Implementation excerpt

```
const geoResponse = await fetch(geocodingUrl);  
if (!geoResponse.ok) throw new Error("City search failed");  
const place = geoData.results[0];  
const weatherResponse = await fetch(forecastUrl);  
if (!weatherResponse.ok) throw new Error("Weather request failed");
```

City names are URL-encoded. Results are inserted as text, and a live status region announces loading and errors.

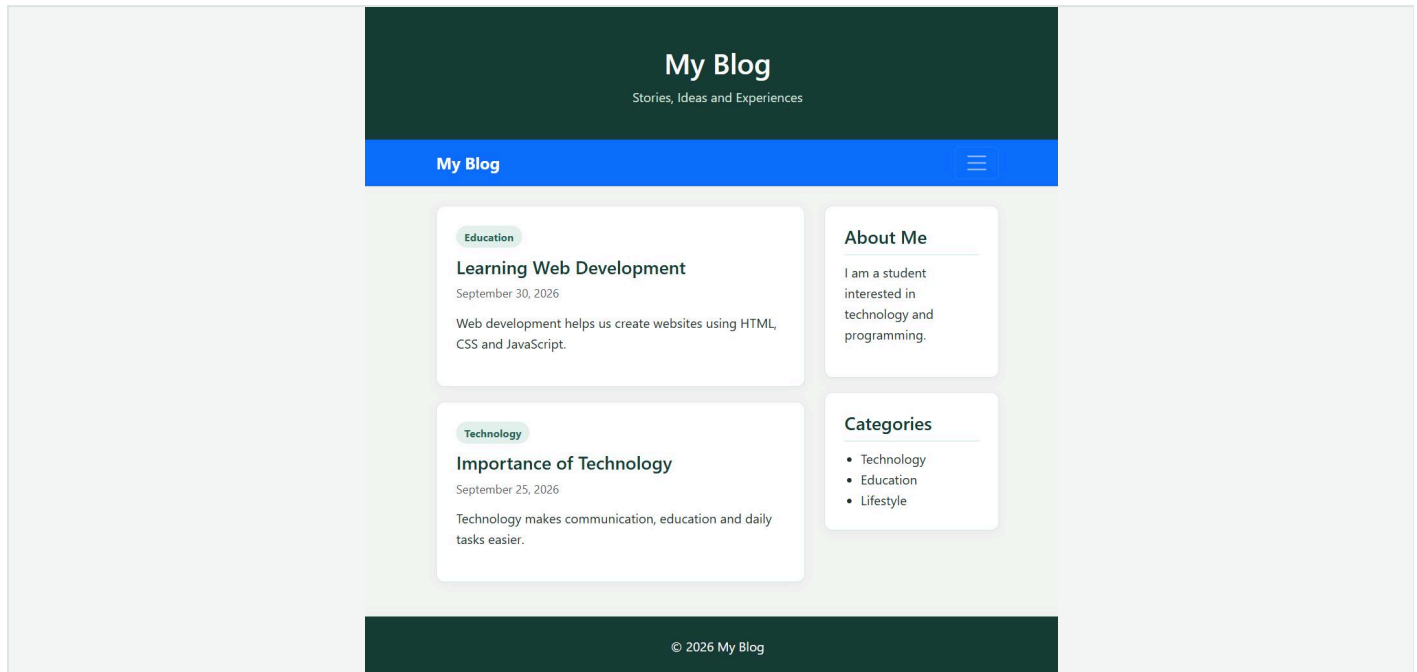
Screenshot reflects the published page at the time of capture. Local source may differ until pushed and deployed.

EXPERIMENT 06 OF 06

Bootstrap Blog

Responsive blog layout using Bootstrap's grid and navbar with post cards, a sidebar, and a footer.

Published page: somyasharma72.github.io/college_experiments/bootstrap_exp6/



Objective and page contents

Present two blog posts in the main column, an About and Categories sidebar, and a footer. The navigation collapses behind a toggle on smaller screens.

Resources: HTML5, Bootstrap 5.3.3 CSS and bundle, and a small custom stylesheet.

Project structure

```
bootstrap_exp6/  
└─ index.html  
(Bootstrap loaded from jsDelivr)
```

Capture: deployed page returned HTTP 200 on 9 Oct 2026.

Implementation excerpt

```
<nav class="navbar navbar-expand-lg navbar-dark">  
  <button class="navbar-toggler"  
    data-bs-toggle="collapse"  
    data-bs-target="#blogNavigation">...</button>  
  ...  
</nav>  
<main class="container">  
  <div class="row">...</div>  
</main>
```

Bootstrap's responsive grid stacks the post column and sidebar on narrow screens.

Screenshot reflects the published page at the time of capture. Local source may differ until pushed and deployed.